

Practising Uncertainty in an Evolving Corpus

Symbolic slippage, compression and continuity in Finite Symbolic Mechanics

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Abstract

A developing corpus cannot hold every distinction with equal clarity in every sentence. Its inherited words and mathematical forms sometimes carry an older licence into an argument being built under new commitments. This note treats such slippage as a question about the evolution of representation, rather than as a collection of isolated mistakes. It uses several crossings among *A Theory of Finite Representation, Part I, Registration of Extent* and *Finite Representation and Interactional Extent* to show how a corpus can retain a recognisable trajectory while revising its vocabulary, operations and bridge rules. The note also proposes a way to locate and measure particular losses of distinction without claiming that one measure exhausts uncertainty.

The shape carried by more than one text

An individual sentence is a narrow opening into a longer construction. A monograph is wider, but it too compresses its sources, experiments, earlier drafts and the movements by which its terms became available. A corpus holds more of that trajectory. Its texts can correct, qualify and reopen one another. They can show a term being used at one stage and reconsidered at another. For work such as Finite Symbolic Mechanics (FSM), which studies how finite interactions become registrations and how registrations enter symbolic trajectories, this is not merely an inconvenience of authorship. It is an instance of the problem under investigation.

The three monographs named above make a provisional sequence. *A Theory of Finite Representation, Part I* states the wider commitments: finite symbol, distinction, uncertainty, provenance and licence, together with the crossings among interactional, symbolic and bridge operations. *Registration of Extent* follows stabilisation towards a detector outcome and asks how a continuing trajectory becomes a compact registration. *Finite Representation and Interactional Extent* makes a further construction by counting what a specified resolution would require of an explicit representation. No one volume is the fully decompressed form of the others. Their relation is a continuing path whose next construction can expose a compression in an earlier one.

The continuity of that path does not demand an unchanging glossary. It lies more deeply in the questions repeatedly asked: where did the distinction occur, what extent did its registration require, what was preserved in the next mark, what uncertainty and history were carried, and which licence permits a symbolic construction to return as a claim about an interaction? When a word slips, those questions give the reader a way to find the intended crossing and, where necessary, alter it.

The pull of an inherited word

The word *physical* is useful in ordinary scientific prose. It can also pull the argument towards a ready-made division: a fully available physical reality on one side and symbols without extent on the other. In a text intended to examine the crossing itself, that habitual division may be introduced before it has been licensed. The phrase “physical basin” in Part I, for example, names the working side of interaction and registration. Read carelessly, it may sound like a basin already known independently of every act of distinction. *Interactional basin* may better indicate the commitment while leaving open what can actually be reconstructed from the registrations.

A sharper example occurs in *Registration of Extent*: “The symbols have no physical volume.” The sentence tries to express the dramatic contraction from detector activity into 0 or 1. Yet an instantiated 0 or 1 occupies a carrier, lasts for a finite interval and can become the start of another interactional trajectory. The distinction encoded by the mark does not reconstruct the detector’s thickness, charge transfer, amplification and timing. Its own finite extent has not vanished. A later formulation can say that the code preserves a selected difference from the interaction while its finite instantiation begins another history. That formulation retains the original insight about compression and repairs the licence under which the word *symbol* is being used.

This is a specific slippage, not a reason to delete every occurrence of *physical*. The word may be needed when reporting a conventional model or discussing an instrument’s material construction. What matters is whether, at a particular crossing, it silently supplies the very separation the argument means to investigate. A developing vocabulary should mark the function of a term in its present sentence, rather than try to erase the history through which the term arrived.

When notation moves between tasks

A related shift occurs without any change of word. In Part I, Δ denotes a finite interactional or measurement window: an extent across which a registration becomes possible. In *Finite Representation and Interactional Extent*, Δ is assigned the width of a spatial interval in a chosen resolution grid. Those uses are related, but they are not identical. If they are silently identified, a sampling choice can seem to have acquired the status of a necessary interactional extent. Subscripts such as Δ_{win} and Δ_{res} would expose the two commitments during a future notation pass. The present corpus already allows the distinction to be recovered by reading each definition in its local construction.

Classical mathematical notation presents a broader case. An integral written with dx can compactly express a useful calculation, including the ledger for costs across an extent. Under the developing FSM commitments, it is an inherited formal operation rather than a completed account of the finite distinctions, overlaps and registration windows from which an empirical result was formed. Its use may be explicitly licensed as a compact stage in a construction. A later finite replacement could unfold the sampling, order, weights, covariance and conditions of passage to a limiting expression. The inherited symbol need not be suppressed before that replacement exists; its role and the work still owed by the reconstruction need to be visible.

The early residual k_{ma} has a similar history. Written beside $F \mid ma$, it can be read as an additional force because the algebraic form invites that reading. Across the wider corpus it can be reconstructed as a mark of unresolved comparison between differently realised trajectories. If a particular residual is later measured in force units and justifies addition, that is an additional bridge. The movement from k_{ma} to a relational comparison record does not make the early notation worthless. It shows what the older form could carry, where it contracted too much, and how a finer distinction became possible.

What can be measured, and what can be lost

The costs discussed in the newer monograph make one part of compression parameterisable. At the stipulated resolution of 10 pm, an explicit interval grid across 1 m has 10^{11} positions, while the ASCII inscription “1 m” occupies three bytes. That contrast measures two chosen descriptions under stated rules. It does not claim that the inscription contains 10^{11} independent observations, or that saying it expends the resources needed to collect them. What it makes visible is the size of one possible decompression relative to a surface representation. Changing the resolution, encoding or required recoverability changes the comparison. This dependence is part of the result.

Other losses can be inspected in a similarly operational way. A detector record can be compared with a binary threshold output: which response shape, timing, calibration and uncertainty fields remain reconstructible from the output and accompanying provenance? A manuscript transcription can be compared with an image of its page: which spacing, marks, damage and alternative readings survive? An astronomical image can be compared with a source catalogue: which pixel relations and segmentation choices remain available after a named object has been entered as a row? In each case the question is attached to actual versions of the record and a declared task. The loss of recoverability may be counted or tested locally even where no single number captures its whole significance.

Uncertainty does not disappear because a symbol omits it. A printed estimate may have no visible uncertainty beside it, but the calibration and fitting histories can still constrain its use. If those histories are unavailable, the omission has changed what can be reconstructed and what claims can be licensed. Compression can therefore make uncertainty *unavailable to a reader* without resolving it in the interaction. This is one reason the apparent certainty of a compact result should be examined through its trajectory rather than through typography alone.

The same concern appears in other representational disciplines. Medical signals become thresholded categories; images become labels; archival objects become catalogue descriptions; multiple measurements become a fitted parameter. These contractions often make work possible. Each also creates a question about the distinctions that remain available for a later decision. The FSM contribution proposed here is to ask that question at the crossing, with attention to extent, uncertainty, provenance and the licence under which the compact result is used.

Practising uncertainty

To practise uncertainty in a corpus is neither to leave every sentence ambiguous nor to pretend a final vocabulary has already been secured. It is to keep a path by which an important ambiguity can be found, named and worked upon. A revised term can point back to the earlier term whose work it took over. A compact equation can state the task for which its operations are licensed. An illustrative interval count can retain the resolution and encoding on which it depends. A newer monograph can revisit an older sentence without making the older trajectory unintelligible.

This also sets a limit on the appeal to a corpus's overall shape. A later chapter cannot make two incompatible claims jointly valid simply by enclosing them in a large volume of writing. The corpus earns its continuity where it makes the movement between them inspectable: what was first meant, what changed, why the change matters and what the new construction permits. Some revisions will be local substitutions; others may change an important bridge or redirect a programme. The reader needs to be able to distinguish them.

For that reason this note is a companion to the corpus rather than an errata sheet. Errata correct a bounded text against a settled intention. Here some intentions are themselves becoming more precise through the writing. The evolving work remains accountable to its recorded slips. Its value is not a promise that every word was perfectly aligned from the beginning. It is the possibility of returning to a compressed symbol, restoring the distinctions needed for a new task, and allowing that reconstruction to alter the next trajectory.